

Multiple Choice

1. **Answer:** C

Options: A) It is used to call procedures with addresses that are farther than 2^{16} bytes away.; B) It cannot return a value.; C) It cannot pass parameters by reference.; D) It cannot call procedures implemented in a different language.

Note: A remote procedure call (RPC) operates over a network, and due to the way data is serialized and transmitted, it only supports passing parameters by value rather than by reference, which would require direct access to memory locations.

2. **Answer:** B

Options: A) `sort('ab')`; B) `sorted('ab')`; C) `"ab".sort()`; D) `1/0`

Note: The expression `'sorted('ab')'` is correct because it uses the built-in `'sorted()'` function in Python to return a list of the characters in the string `'ab'` sorted in ascending order, which is a valid operation in Python 3.5.

3. **Answer:** C

Options: A) `{1}`; B) `{1,2,2,3,4}`; C) `{1,2,3,4}`; D) `{4,3,2,2,1}`

Note: The correct answer is `{1,2,3,4}` because the `'set()'` function in Python removes duplicate values from the list and returns a collection of unique elements.

4. **Answer:** A

Options: A) `xyz`; B) `xy`; C) `xxzy`; D) `xxxxy`

Note: The string `"xyz"` is not generated by the given grammar because it does not conform to the production rules, specifically lacking the necessary structure that incorporates the non-terminal symbols B, C, or D as required by the grammar.

5. **Answer:** A

Options: A) Reference counting is well suited for reclaiming cyclic structures.; B) Reference counting incurs additional space overhead for each memory cell.; C) Reference counting is an alternative to mark-and-sweep garbage collection.; D) Reference counting need not keep track of which cells point to other cells.

Note: Reference counting is not well suited for reclaiming cyclic structures because it cannot effectively detect and reclaim memory used by objects that reference each other, resulting in memory leaks.

True / False

6. **Answer:** False

Options: A) True; B) False

Note: The statement is false because the expression `['Hi!'] * 4` is valid syntax in Python 3 and creates a list containing four copies of the string `"Hi!"`.

7. **Answer:** False

Options: A) True; B) False

Note: Identifier length in programming languages is best specified using regular expressions rather than context-free grammars because context-free grammars can only define structures based on nested and recursive patterns, while identifier length constraints are linear and can be expressed more simply with regular expressions.

8. **Answer:** False

Options: A) True; B) False

Note: The statement is false because the number of distinct functions from a finite set A with m elements to a finite set B with n elements is n^m , not $n!/(n - m)!$, which incorrectly suggests a one-to-one correspondence between the elements of A and B.

9. **Answer:** False

Options: A) True; B) False

Note: Church's thesis, also known as the Church-Turing thesis, was not proven by Alan Turing but rather is a hypothesis about the nature of computable functions that emerged from the work of both Alonzo Church and Turing independently, establishing that their models of computation were equivalent.

10. **Answer:** False

Options: A) True; B) False

Note: Interrupts are mechanisms that signal the CPU to pause its current operations to address an event, whereas data channels, such as Direct Memory Access (DMA), are dedicated pathways that allow devices to transfer data directly to and from memory without involving the CPU, highlighting their distinct roles in data transfer.

Long Form Response

11. **Answer:** `def is_Perfect_Square(n) : | return True | return False`

Note: correct as it defines a function that checks if a number is a perfect square by returning True or False based on the logic implemented within the function, although the actual logic to determine whether 'n' is a perfect square is missing.

12. **Answer:** `def parallelogram_perimeter(b,h): | return perimeter`

Note: correct because it outlines the basic structure of a function that is intended to calculate the perimeter of a parallelogram, although it lacks the actual implementation of the formula, which is 2 times the sum of the base and height.

End of Answer Key